

A Comprehensive Introduction to Semisimple Lie Algebras for Theoretical Physicists

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Introduction

Lie algebras are the natural language of continuous symmetries. In quantum mechanics, the angular momentum operators L_x, L_y, L_z close under commutation, forming the Lie algebra $\mathfrak{so}(3)$. In particle physics, the Gell-Mann matrices satisfy the commutation relations of $\mathfrak{su}(3)$, the algebra of the strong force. More generally, any continuous symmetry group of a physical system gives rise to a Lie algebra of infinitesimal transformations, whose structure deeply constrains the possible spectra, interactions, and dynamics.

This text aims to provide theoretical physicists with a self-contained, rigorous yet physically motivated exposition of semisimple Lie algebras and their classification. We will follow the notation and logical development of Brian Hall's textbook [1], filling in physical insights, examples, and applications at every stage. By the end, the reader will understand why the possible simple Lie algebras are precisely $A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4, G_2$, how this classification underpins the structure of the Standard Model, and how the roots and weights organise particle multiplets.

We assume the reader is comfortable with linear algebra and knows a little quantum mechanics. No prior exposure to abstract Lie theory is required.

1 Basic Concepts: Solvability, Semisimplicity, and the Radical

Definition 1.1. A **Lie algebra** over $\mathbb{C}$ is a vector space $\mathfrak{g}$ with a bilinear bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ that is alternating ($[X, Y] = -[Y, X]$) and satisfies the Jacobi identity

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

Physicists encounter Lie algebras as the commutator algebras of observables. For instance, the angular momentum operators satisfy $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$, which is a real Lie algebra (after factoring out i). We will work over the complex numbers for mathematical convenience; real forms can then be studied by restricting to suitable real subalgebras.

The **derived series** is $\mathfrak{g}^{(0)} = \mathfrak{g}$, $\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$, and the **lower central series** is $\mathfrak{g}^{[0]} = \mathfrak{g}$, $\mathfrak{g}^{[k+1]} = [\mathfrak{g}, \mathfrak{g}^{[k]}]$. $\mathfrak{g}$ is **solvable** if $\mathfrak{g}^{(n)} = 0$ for some n , and **nilpotent** if $\mathfrak{g}^{[n]} = 0$. These notions formalise the idea of a Lie algebra that can be “integrated” step by step. In classical mechanics, solvable Lie algebras appear as algebras of canonical transformations for which the equations of motion can be solved by quadratures. In quantum mechanics, a set of operators whose commutators become successively simpler (nilpotent) generates a Heisenberg–Weyl type algebra.

Example 1.2 (Differential equations, motivation). A finite-dimensional Lie algebra of vector fields on a manifold M with $[V_i, V_j] = \sum c_{ij}^k V_k$ is completely integrable (in the sense of Frobenius) iff it is a subalgebra. If the Lie algebra is solvable, the corresponding linear ordinary differential equations can be integrated by successive quadratures — Lie’s Theorem below guarantees a common eigenvector which reduces the order. This historical observation ties the algebraic notion of solvability to the solvability of differential equations by elementary functions.

Definition 1.3. The **radical** $\text{rad}(\mathfrak{g})$ is the unique maximal solvable ideal of $\mathfrak{g}$. $\mathfrak{g}$ is **semisimple** if $\text{rad}(\mathfrak{g}) = 0$, and **simple** if it has no proper nonzero ideals and $\dim \mathfrak{g} > 1$.

In physics, semisimple algebras are those for which all solvable “tails” have been stripped away; they are the building blocks of symmetry groups. The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ is reductive (direct sum of a semisimple part $SU(3) \times SU(2)$ and an abelian $U(1)$). The classification of semisimple algebras thus directly determines possible gauge groups.

2 Lie’s and Engel’s Theorems (Toolbox)

The two fundamental theorems that govern solvable and nilpotent Lie algebras are stated here for completeness; their proofs can be found in [1].

Theorem 2.1 (Lie’s Theorem). *If $\mathfrak{g}$ is a solvable Lie algebra and $\rho : \mathfrak{g} \rightarrow \text{End}(V)$ is a representation on a finite-dimensional complex vector space $V \neq 0$, then there exists a nonzero $v \in V$ that is a common eigenvector for all $\rho(X)$.*

The proof inducts on $\dim \mathfrak{g}$: pick a codimension-one ideal $\mathfrak{a}$, use induction to get an eigenvector for $\mathfrak{a}$, then study the cyclic space generated by a complementary element Y . The upshot is

$[\mathfrak{g}, \mathfrak{g}]$ acts by nilpotent operators in any representation.

Physically, this means that if we have a set of operators that generate a solvable Lie algebra, we can find a basis in which they are all upper-triangular. This is the algebraic reason why certain quantum-mechanical problems (e.g., the harmonic oscillator) admit a ladder of states that terminates.

Theorem 2.2 (Engel’s Theorem). *A Lie algebra $\mathfrak{g}$ is nilpotent iff $\text{ad } X$ is nilpotent for every $X \in \mathfrak{g}$.*

The nontrivial direction follows from the existence of a nonzero central element when all $\text{ad } X$ are nilpotent, and an induction on dimension.

3 The Killing Form and Cartan’s Criteria

Definition 3.1. The **Killing form** on $\mathfrak{g}$ is the symmetric bilinear form

$$K(X, Y) = \text{tr}(\text{ad } X \circ \text{ad } Y).$$

It is *invariant*: $K([X, Y], Z) = K(X, [Y, Z])$.

For matrix Lie algebras (subalgebras of $\mathfrak{gl}(n, \mathbb{C})$), the Killing form is just (up to a constant) the trace of the product: $\text{tr}(XY)$. For $\mathfrak{su}(2)$, $\text{tr}(\sigma_a \sigma_b) = 2\delta_{ab}$, which is proportional to the usual Euclidean inner product. This metric structure is fundamental: in gauge theories, the Killing form is used to write kinetic terms for gauge fields, $-\frac{1}{4g^2} K(F_{\mu\nu}, F^{\mu\nu})$.

Theorem 3.2 (Cartan’s First Criterion). *$\mathfrak{g}$ is solvable iff $K(X, Y) = 0$ for all $X \in \mathfrak{g}$ and $Y \in [\mathfrak{g}, \mathfrak{g}]$.*

Sketch. If $\mathfrak{g}$ is solvable, Lie's theorem applied to the adjoint representation triangularizes all $\text{ad } X$; for $Y \in [\mathfrak{g}, \mathfrak{g}]$, $\text{ad } Y$ is strictly upper-triangular, so its product with any $\text{ad } X$ has zero trace. Conversely, if the condition holds, one shows that $\mathfrak{g}$ cannot contain a subalgebra isomorphic to $\mathfrak{sl}_2(\mathbb{C})$ (whose Killing form is nondegenerate), forcing solvability. Details are in [1], Thm. 5.32. $\square$

Theorem 3.3 (Cartan's Second Criterion). *$\mathfrak{g}$ is semisimple iff K is nondegenerate.*

Proof. Let $\mathfrak{r} = \{X \in \mathfrak{g} \mid K(X, \mathfrak{g}) = 0\}$ be the radical of the form. Invariance implies $\mathfrak{r}$ is an ideal; Cartan's first criterion shows $\mathfrak{r}$ is solvable, hence $\mathfrak{r} \subseteq \text{rad}(\mathfrak{g})$. If $\mathfrak{g}$ is semisimple, $\text{rad}(\mathfrak{g}) = 0$, so $\mathfrak{r} = 0$ and K is nondegenerate. Conversely, if K is nondegenerate, any solvable ideal $\mathfrak{a}$ satisfies $K(\mathfrak{a}, \mathfrak{g}) = 0$ (Lie's theorem again), so $\mathfrak{a} = 0$; thus $\text{rad}(\mathfrak{g}) = 0$. $\square$

Corollary 3.4 (Decomposition into simples). *A semisimple Lie algebra $\mathfrak{g}$ decomposes as a direct sum $\mathfrak{g} = \bigoplus \mathfrak{g}_i$ of simple ideals, orthogonal with respect to K .*

Proof. For any ideal $\mathfrak{a}$, its orthogonal complement $\mathfrak{a}^\perp$ is also an ideal and $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$ when $K|_{\mathfrak{a}}$ is nondegenerate. Iterating yields the decomposition. $\square$

From now on we assume $\mathfrak{g}$ is a ****complex semisimple Lie algebra****. The nondegenerate Killing form will be our main tool. Physically, this tells us that any semisimple gauge algebra splits into mutually non-interacting simple factors, each with its own coupling constant.

4 Cartan Subalgebras, Roots, and the Inner Product

4.1 Cartan subalgebras

Definition 4.1. A **Cartan subalgebra** $\mathfrak{h}$ of $\mathfrak{g}$ is a maximal abelian subalgebra consisting of ad-diagonalizable (semisimple) elements.

In physical terms, $\mathfrak{h}$ corresponds to a maximal set of commuting observables that can be simultaneously diagonalized. For $SU(2)$, $\mathfrak{h}$ is the one-dimensional algebra generated by J_3 ; for $SU(3)$, $\mathfrak{h}$ is the two-dimensional algebra spanned by the two diagonal Gell-Mann matrices λ_3 and λ_8 .

Cartan subalgebras always exist: take a *regular* element $H_0 \in \mathfrak{g}$, i.e., one for which the nilspace of $\text{ad } H_0$ has minimal dimension. Then the generalized nullspace (the span of generalized eigenvectors with eigenvalue 0) is a Cartan subalgebra. All Cartan subalgebras are conjugate by inner automorphisms, and their common dimension is the **rank** of $\mathfrak{g}$. The rank is the number of independent additive quantum numbers (like isospin I_3 , hypercharge Y) that can be simultaneously assigned to particles in a representation.

Since $\mathfrak{h}$ consists of commuting semisimple operators, they can be simultaneously diagonalized:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \quad (4.1)$$

where $R \subset \mathfrak{h}^* \setminus \{0\}$ is a finite set of **roots**, and

$$\mathfrak{g}_\alpha = \{X \in \mathfrak{g} \mid [H, X] = \alpha(H)X \ \forall H \in \mathfrak{h}\}.$$

The finiteness of R follows because $\mathfrak{g}$ is finite-dimensional. The decomposition is orthogonal with respect to the Killing form: $K(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$ unless $\alpha + \beta = 0$.

The restriction of K to $\mathfrak{h}$ is nondegenerate (otherwise an $H \in \mathfrak{h}$ orthogonal to $\mathfrak{h}$ would be orthogonal to all $\mathfrak{g}_\alpha$ as well, forcing $H = 0$ by nondegeneracy of K on $\mathfrak{g}$). Thus we have an isomorphism

$$\phi : \mathfrak{h} \longrightarrow \mathfrak{h}^*, \quad H \mapsto (H' \mapsto K(H, H')).$$

For $\lambda \in \mathfrak{h}^*$, denote by t_λ the unique element of $\mathfrak{h}$ with $\lambda(H) = K(t_\lambda, H)$ for all H .

4.2 Inner product on the root space

Define a bilinear form on $\mathfrak{h}^*$ by

$$(\lambda, \mu) := K(t_\lambda, t_\mu).$$

Let $E = \text{span}_{\mathbb{R}} R \subset \mathfrak{h}^*$; this is a real form of $\mathfrak{h}^*$ ($\mathfrak{h}^* = E \oplus iE$).

Proposition 4.2. *$(\cdot, \cdot)$ restricted to E is a real-valued, positive definite inner product.*

Proof. For any $H = t_\lambda$ with $\lambda \in E$, we compute using (4.1) that

$$K(H, H) = \text{tr}((\text{ad } H)^2) = \sum_{\alpha \in R} \alpha(H)^2.$$

Indeed, on $\mathfrak{h}$, $\text{ad } H = 0$; on each one-dimensional space $\mathfrak{g}_\alpha$, $\text{ad } H$ acts as multiplication by $\alpha(H)$. Hence the trace of $(\text{ad } H)^2$ is the sum over all roots of $\alpha(H)^2$. (Note that we will prove $\dim \mathfrak{g}_\alpha = 1$ shortly, but this formula holds even before that if we count each root with multiplicity.) Thus

$$(\lambda, \lambda) = K(t_\lambda, t_\lambda) = \sum_{\alpha \in R} \alpha(t_\lambda)^2 = \sum_{\alpha \in R} (\lambda, \alpha)^2 \geq 0.$$

If $(\lambda, \lambda) = 0$, then $(\lambda, \alpha) = 0$ for all $\alpha \in R$, so λ is orthogonal to the entire span of the roots, i.e., to E . Since $(\cdot, \cdot)$ is nondegenerate on $\mathfrak{h}^*$, this forces $\lambda = 0$. Hence the form is positive definite. $\square$

Henceforth we write $\langle \lambda, \mu \rangle = (\lambda, \mu)$ for the inner product on E . This Euclidean structure is what allows us to draw root diagrams and measure angles and lengths. Physically, the inner product gives the metric on the space of quantum numbers.

4.3 $\mathfrak{sl}_2$ subalgebras and the fundamental root properties

Fix a root $\alpha \in R$. Because K pairs $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$ nondegenerately (if α is a root, so is $-\alpha$, and otherwise $K(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$ unless $\beta = -\alpha$), we can pick $X_\alpha \in \mathfrak{g}_\alpha$, $Y_\alpha \in \mathfrak{g}_{-\alpha}$ such that $K(X_\alpha, Y_\alpha) = 1$. A direct calculation shows

$$[X_\alpha, Y_\alpha] = -t_\alpha.$$

Indeed, for any $H \in \mathfrak{h}$, using invariance,

$$K([X_\alpha, Y_\alpha], H) = K(X_\alpha, [Y_\alpha, H]) = -K(X_\alpha, \alpha(H)Y_\alpha) = -\alpha(H)K(X_\alpha, Y_\alpha) = -\alpha(H).$$

On the other hand, $K(t_\alpha, H) = \alpha(H)$ by definition. Since K on $\mathfrak{h}$ is nondegenerate, we conclude $[X_\alpha, Y_\alpha] = -t_\alpha$.

Now rescale:

$$E_\alpha = \frac{2}{\langle \alpha, \alpha \rangle} X_\alpha, \quad F_\alpha = Y_\alpha, \quad H_\alpha = \frac{2}{\langle \alpha, \alpha \rangle} t_\alpha.$$

Using $K(t_\alpha, t_\alpha) = \langle \alpha, \alpha \rangle$, one verifies the $\mathfrak{sl}_2(\mathbb{C})$ commutation relations:

$$[H_\alpha, E_\alpha] = 2E_\alpha, \quad [H_\alpha, F_\alpha] = -2F_\alpha, \quad [E_\alpha, F_\alpha] = H_\alpha.$$

Thus for each $\alpha \in R$, we obtain a subalgebra $\mathfrak{s}_\alpha \cong \mathfrak{sl}_2(\mathbb{C})$.

The adjoint action of this $\mathfrak{s}_\alpha$ on the whole $\mathfrak{g}$ makes $\mathfrak{g}$ a (finite-dimensional) $\mathfrak{sl}_2(\mathbb{C})$ -module. The representation theory of $\mathfrak{sl}_2(\mathbb{C})$ is elementary and well-known to physicists from angular momentum: in any finite-dimensional representation,

- All eigenvalues of H_α are integers.
- The eigenspaces (weight spaces) are finite-dimensional.
- If v is an eigenvector of H_α with eigenvalue λ (called a weight vector), then $E_\alpha v$ (if nonzero) has eigenvalue $\lambda + 2$, and $F_\alpha v$ has eigenvalue $\lambda - 2$.
- The set of eigenvalues (weights) of H_α forms an unbroken string $\lambda, \lambda - 2, \dots, \lambda - 2k$ where k is the length of the string, symmetric about zero. In particular, the highest and lowest weights are negatives of each other.

These facts are proven by standard ladder operator arguments, completely analogous to those for the harmonic oscillator or angular momentum.

We now apply this to $\mathfrak{g}$ itself. Consider the α -string through a root β :

$$\mathfrak{g}(\beta, \alpha) = \bigoplus_{k \in \mathbb{Z}} \mathfrak{g}_{\beta + k\alpha}.$$

This is an $\mathfrak{sl}_2$ -submodule of $\mathfrak{g}$, because $E_\alpha \in \mathfrak{g}_\alpha$, $F_\alpha \in \mathfrak{g}_{-\alpha}$ and $H_\alpha \in \mathfrak{h}$ preserve the root spaces as indicated.

On $\mathfrak{g}_{\beta + k\alpha}$, H_α acts as the scalar

$$\frac{2}{\langle \alpha, \alpha \rangle} (\beta + k\alpha)(t_\alpha) = \frac{2}{\langle \alpha, \alpha \rangle} (\langle \beta, \alpha \rangle + k\langle \alpha, \alpha \rangle) = 2 \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} + 2k.$$

Since these numbers must be integers (they are eigenvalues of H_α in a finite-dimensional module), we deduce the first fundamental fact:

$$n_{\beta\alpha} := 2 \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} \in \mathbb{Z}. \quad (4.2)$$

These integers are called **Cartan integers**.

Moreover, the set of k for which $\mathfrak{g}_{\beta+k\alpha} \neq 0$ is an unbroken string from $k = -p$ to $k = q$, where $p, q \geq 0$. The highest weight in this string is $2 \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} + 2q$, and the lowest is $-2 \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} - 2p$. By symmetry,

$$2 \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} + 2q = -(-2 \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} - 2p) \implies p - q = 2 \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle}. \quad (4.3)$$

Now consider the special case $\beta = \alpha$. The string $\mathfrak{g}_{\alpha+k\alpha}$ contains $\mathfrak{g}_\alpha$ at $k = 0$. Applying E_α to $\mathfrak{g}_\alpha$ gives an element of $\mathfrak{g}_{2\alpha}$, but we claim 2α is not a root. Indeed, if 2α were a root, the H_α weight on $\mathfrak{g}_{2\alpha}$ would be 4. The α -string through 0 (i.e., taking $\beta = 0$, though 0 is not a root but we can consider the $\mathfrak{h}$ part as $\mathfrak{g}_0$) has weights that must be symmetric integers. The action of H_α on $\mathfrak{h}$ is zero, so the string through 0 has weights $\dots, -2, 0, 2, \dots$. If $\mathfrak{g}_{2\alpha} \neq 0$, the weight 4 would appear, contradicting the fact that in the string through 0 the highest weight cannot exceed the maximal eigenvalue, which is bounded by the dimension. More rigorously, the $\mathfrak{sl}_2$ submodule generated by $\mathfrak{g}_\alpha$ (which is a highest weight vector because $E_\alpha X_\alpha \in \mathfrak{g}_{2\alpha} = 0$ by the root property) must have highest weight 2 (since H_α acts as 2 on $\mathfrak{g}_\alpha$). The dimension of this irreducible module is 3, with weights $\{-2, 0, 2\}$. The weight space of 2 is $\mathfrak{g}_\alpha$ (one-dimensional), of 0 is $\mathfrak{h}$ (or a subspace thereof), and of -2 is $\mathfrak{g}_{-\alpha}$. Hence $\dim \mathfrak{g}_\alpha = 1$, and no other multiples of α ($\pm 2\alpha$, etc.) can appear. This completes the proof that

1. $\dim \mathfrak{g}_\alpha = 1$,
2. the only multiples of α in R are $\pm\alpha$.

Finally, using the string condition (4.3) with p, q , we compute the reflection s_α on β :

$$s_\alpha(\beta) = \beta - 2 \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha = \beta - (q - p)\alpha.$$

Because the string is symmetric, $\beta - (q - p)\alpha$ is exactly $\beta - m\alpha$ where $m = q - p$, and one checks that this belongs to the string (it is the element at position $-p$ after shifting). Hence $s_\alpha(\beta) \in R$. Thus

3. $s_\alpha(R) = R$ for all $\alpha \in R$.

We have now established all the fundamental properties of roots using only the existence of the Cartan subalgebra, the nondegenerate Killing form, and the representation theory of $\mathfrak{sl}_2$. These properties will be crystallized into the abstract definition of a root system.

4.4 The Weyl group

Definition 4.3. The **Weyl group** $\mathcal{W}$ is the subgroup of the orthogonal group $O(E)$ generated by the reflections $\{s_\alpha \mid \alpha \in R\}$.

Each s_α permutes the finite set R , and R spans E ; hence the map $\mathcal{W} \rightarrow \text{Perm}(R)$ is injective, and $\mathcal{W}$ is a finite group (a subgroup of the symmetric group on $|R|$ elements). In physical applications, the Weyl group is the symmetry of the weight diagram; it organizes weights of representations into orbits.

5 Root Systems and their Geometry

The properties we have derived are so restrictive and elegant that we now abstract them, allowing us to classify all possibilities independently of the original Lie algebra.

Definition 5.1. A **reduced root system** is a finite set R of nonzero vectors in a Euclidean space E (with inner product $\langle \cdot, \cdot \rangle$) satisfying:

- (R1) R spans E .
- (R2) For each $\alpha \in R$, the only scalar multiples of α that belong to R are $\pm\alpha$.
- (R3) For each $\alpha \in R$, the reflection $s_\alpha(\lambda) = \lambda - 2\frac{\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha$ leaves R invariant.
- (R4) For any $\alpha, \beta \in R$, $n_{\beta\alpha} = 2\frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} \in \mathbb{Z}$.

We have proved that the roots of a semisimple Lie algebra form a reduced root system. Conversely, every reduced root system arises from some semisimple Lie algebra (Serre's theorem, Section 7). Hence classifying semisimple Lie algebras is equivalent to classifying reduced root systems.

A root system is **irreducible** if it cannot be expressed as the orthogonal disjoint union of two nonempty subsets. The Lie algebra is simple iff its root system is irreducible.

5.1 Simple roots and bases

To work with a root system, we choose a direction.

Definition 5.2. A subset $\Delta \subset R$ is a **base** (or system of simple roots) if

- Δ is a basis of E ,
- every $\beta \in R$ can be written as $\beta = \sum_{\alpha \in \Delta} c_\alpha \alpha$ with integer coefficients c_α that are either all ≥ 0 (positive roots) or all ≤ 0 (negative roots).

Proposition 5.3. *A base always exists.*

Proof. Pick a vector $v \in E$ that is *regular*, i.e., $\langle v, \alpha \rangle \neq 0$ for all $\alpha \in R$. (Such vectors exist because the hyperplanes $\alpha^\perp$ do not cover E .) Declare a root α to be *positive* if $\langle v, \alpha \rangle > 0$, and negative otherwise. An element of R^+ (the set of positive roots) is *ndecomposable* if it cannot be written as a sum of two other positive roots. One proves that the set Δ of all indecomposable positive roots is a base. The proof uses the finite generation property and integrality. See [1], Thm. 8.16 for details. $\square$

The Weyl group acts simply transitively on the set of all bases.

Given a base $\Delta = \{\alpha_1, \dots, \alpha_r\}$ (r is the rank), the **Cartan matrix** $A = (A_{ij})$ is defined by

$$A_{ij} = 2 \frac{\langle \alpha_j, \alpha_i \rangle}{\langle \alpha_i, \alpha_i \rangle} = n_{\alpha_j \alpha_i}.$$

All information about the root system is encoded in this integer matrix. In particular,

- $A_{ii} = 2$ for all i .
- For $i \neq j$, $A_{ij} \in \{0, -1, -2, -3\}$ because $A_{ij}A_{ji} = 4 \cos^2 \theta_{ij}$ with θ_{ij} the angle between α_i and α_j . Since the roots are not scalar multiples, $A_{ij}A_{ji}$ can only be 0, 1, 2, 3.
- $A_{ij} = 0 \iff A_{ji} = 0$.
- The Cartan matrix is positive definite ($\det A > 0$), which follows from the positive definiteness of the inner product.

5.2 Dynkin diagrams

The Cartan matrix is graphically represented by the **Dynkin diagram**:

- Nodes correspond to the simple roots $\alpha_1, \dots, \alpha_r$.
- Two nodes $i \neq j$ are connected by $A_{ij}A_{ji}$ edges (0,1,2,3).
- If a multiple edge occurs, we draw an arrow pointing from the longer root to the shorter root (i.e., from the node with larger $\langle \alpha_i, \alpha_i \rangle$ to the one with smaller).

The root system is irreducible iff the Dynkin diagram is connected. The classification of irreducible root systems reduces to the classification of connected Dynkin diagrams.

5.3 Classification theorem

Theorem 5.4 (Cartan–Killing classification). *Every irreducible reduced root system is isomorphic to exactly one of the following families, with Dynkin diagrams as shown:*

Type	Dynkin diagram
A_n ($n \geq 1$)	$\circ - \circ - \cdots - \circ$ (n nodes)
B_n ($n \geq 2$)	$\circ - \circ - \cdots - \circ \Rightarrow \circ$
C_n ($n \geq 3$)	$\circ - \circ - \cdots - \circ \Leftarrow \circ$
D_n ($n \geq 4$)	$\circ - \circ - \cdots - \circ$ with an extra node attached to the penultimate
E_6, E_7, E_8	exceptional branching diagrams
F_4	$\circ - \circ \Rightarrow \circ - \circ$
G_2	$\circ \Rightarrow \circ$ (triple edge)

Proof outline. The proof is a beautiful exercise in Euclidean geometry.

1. Show that the Coxeter graph (ignore arrows) is a tree. If there were a cycle, the sum of angles around the cycle would exceed 2π , forcing the Cartan matrix to be non-positive definite, contradiction.
2. Bound the degree of any node. Since the simple roots are linearly independent, a node connected to k other nodes would imply that these k vectors have a nontrivial null linear combination, again using positive definiteness. One finds that a node can have at most three edges, and certain patterns (like a triple edge plus another edge) are forbidden.
3. These constraints eliminate all graphs except the known families A_n , B_n ($= C_n$), D_n , E_6, E_7, E_8 , F_4 , and G_2 .
4. Finally, the arrow directions are determined by the relative root lengths. For B_n and C_n , which have a double edge, the arrow distinguishes between the algebra where the short root is at the end (B_n) or the long root is at the end (C_n). For F_4 and G_2 , the arrows are fixed by consistency.

A full exposition of this combinatorial argument can be found in [1], Ch. 8. $\square$

Consequently, any complex simple Lie algebra belongs to one of the types A_n , B_n , C_n , D_n , E_6 , E_7 , E_8 , F_4 , G_2 . A semisimple Lie algebra is a direct sum of simple such pieces.

Physical remark 5.5. In physics, the A_n series corresponds to $\mathfrak{su}(n+1)$; B_n to $\mathfrak{so}(2n+1)$; C_n to $\mathfrak{sp}(2n)$; D_n to $\mathfrak{so}(2n)$. The exceptional groups E_6, E_7, E_8 appear in Grand Unified Theories (GUTs) and string theory; F_4 and G_2 arise in certain supergravity and M-theory compactifications.

6 Representations and Weights (A Physicist's View)

For completeness, we briefly sketch how representation theory ties into the above. A finite-dimensional representation of $\mathfrak{g}$ on a vector space V is a Lie algebra homomorphism $\pi : \mathfrak{g} \rightarrow \text{End}(V)$. When restricted to a Cartan subalgebra $\mathfrak{h}$, the operators $\pi(H)$ for $H \in \mathfrak{h}$ are simultaneously diagonalizable (by preservation of the Killing form, one can define a Cartan involution, etc.). Thus V decomposes into **weight spaces**

$$V = \bigoplus_{\mu \in \Phi} V_{\mu}, \quad V_{\mu} = \{v \in V \mid \pi(H)v = \mu(H)v\}.$$

The weights μ are elements of $\mathfrak{h}^*$, and their inner product with the roots governs which root operators $\pi(X_{\alpha})$ map V_{μ} to $V_{\mu+\alpha}$.

The **highest weight** of an irreducible representation is a weight λ such that $V_{\lambda+\alpha} = 0$ for all positive roots α . The representation is uniquely determined by this highest weight, which must be **dominant integral** (i.e., $2\langle\lambda, \alpha\rangle/\langle\alpha, \alpha\rangle$ is a non-negative integer for every simple root α). The Weyl group orbit of the highest weight generates all weights of the representation, and the multiplicity patterns are given by the celebrated Weyl character formula.

This machinery is exactly what is used to classify elementary particles in GUTs: the known particles are assembled into an irreducible representation of the gauge algebra, and the weights (quantum numbers) are computed using the simple roots of the Dynkin diagram.

Example 6.1 ($SU(3)$ quark model). For A_2 ($\mathfrak{su}(3)$), the rank is 2. The simple roots are α_1, α_2 with $\angle = 120^\circ$ and equal lengths. The fundamental representation **3** has highest weight $\Lambda_1 = \frac{2\alpha_1 + \alpha_2}{3}$, while the antifundamental $\bar{\mathbf{3}}$ has highest weight $\Lambda_2 = \frac{\alpha_1 + 2\alpha_2}{3}$. The adjoint representation **8** consists of the roots themselves plus the two zero weights from $\mathfrak{h}$. The meson and baryon octets, decuplets, etc., are all explained by tensor products of these basic representations, governed by the Clebsch–Gordan coefficients derived from the root and weight geometry.

7 Reconstruction: Serre's Theorem

To close the circle, we state that every abstract root system (or Cartan matrix) indeed comes from a Lie algebra.

Theorem 7.1 (Serre's presentation). *Let $A = (A_{ij})$ be an $r \times r$ Cartan matrix of finite type (i.e., all principal minors positive). Define $\mathfrak{g}(A)$ as the Lie algebra generated by e_i, f_i, h_i ($i = 1, \dots, r$) subject to the relations*

$$[h_i, h_j] = 0, \quad [e_i, f_j] = \delta_{ij} h_i, \quad [h_i, e_j] = A_{ij} e_j, \quad [h_i, f_j] = -A_{ij} f_j,$$

and the Serre relations

$$(\operatorname{ad} e_i)^{1-A_{ij}}(e_j) = 0, \quad (\operatorname{ad} f_i)^{1-A_{ij}}(f_j) = 0 \quad (i \neq j).$$

Then $\mathfrak{g}(A)$ is a finite-dimensional complex semisimple Lie algebra whose Cartan matrix (with respect to $\mathfrak{h} = \operatorname{span}\{h_i\}$) is exactly A .

Physical remark 7.2. In physics, the generators e_i, f_i, h_i correspond to raising, lowering, and diagonal (Cartan) operators, respectively. The Serre relations are the algebraic expression of the fact that repeated action of raising/lowering operators eventually vanishes inside a finite-dimensional representation — ultimately a consequence of the finite number of states in a multiplet.

Thus the classification of complex simple Lie algebras is perfectly mirrored by the classification of irreducible Dynkin diagrams, and the whole theory is completely self-contained.

Conclusion

We have developed the complete classification of semisimple Lie algebras, starting from first principles and building up to the Dynkin diagrams and Serre's reconstruction. The route goes through solvability, the Killing form, Cartan subalgebras, the $\mathfrak{sl}_2$ subalgebra machinery, root systems, and finally the combinatorial classification. This framework not only organises all possible continuous symmetries but also provides the computational tools (roots, weights, Weyl group) necessary to construct and analyze their representations—the very vocabulary in which much of modern theoretical physics is written.

References

- [1] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed., GTM 222, Springer, 2015.